VOICEOPS 2026 / CAPSTONE REPORT

A support agent that can show its work

ZentroMart voice support agent · Voxentra team · 25 October 2026

We built a customer-support agent that answers from ZentroMart's policy documents, uses the supplied tables and mock tools for order-specific facts, and speaks replies through Gnani. The main design choice was to keep policy interpretation, private order data, and speech handling in separate parts of the system. That makes it easier to see what evidence supports an answer and where a human still needs to decide.

How the four steps fit together

STEP	WHAT WE BUILT
1 · Baseline	A concise Gemini support prompt, evaluated before adding policy retrieval.
2 · Grounding	Policy Markdown is indexed by section ID and ranked with hybrid retrieval. Current policies take precedence over guides, help pages, FAQs, and marketing. Product, pincode, holiday, and order facts come from structured data or the supplied mock tools.
3 · Conversation	A stateful controller carries issue and order context across turns. It verifies callers before sharing private order details, allows one retry after a failed verification, and creates a ticket before confirming a handover.
4 · Voice	Gnani handles both speech recognition and speech synthesis, with the named `timbre-v2.5` TTS model. The same conversation agent processes the transcript; voice formatting prepares dates, amounts, and order numbers to be spoken clearly.

A narrow safety boundary

The agent uses only the synthetic capstone data. It does not reveal order details until the mock lookup verifies the caller with the registered phone's last four digits or the delivery pincode. Verification values are redacted from saved conversation records. A handover is described as complete only after ticket creation succeeds. Gnani is the only speech provider; Gemini is used for text generation and embeddings.

What the practice results show

Grounding made the clearest difference. On the same 52 practice questions, the RAG agent's local pass rate was 51.9 percentage points higher than the prompt-only baseline. The speech-like transcript set scored higher again, though it is a text exercise rather than a measure of speech recognition accuracy.

RUN	CASES	PASS RATE	FACT COVERAGE	OTHER
Step 1 · Prompt baseline	52	**30.8%**	24.1%	No retrieval
Step 2 · RAG	52	**82.7%**	89.7%	92.3% decision accuracy
Step 4 · ASR-style text v2	52	**92.3%**	91.4%	98.1% decision accuracy

The local retrieval run found labeled evidence in 47 of 47 evidence-bearing questions at top five (Hit@5 100%). Mean reciprocal rank was 0.952 and evidence recall was 0.805. The full practice conversation set passed 96 of 96 local string and routing checks across 23 conversations. A targeted tool-argument review passed all four turns in one failed-verification scenario.

These are practice results from the project's approximate local scorer, not the organizers' hidden evaluation. The scorer checks expected facts, decisions, and evidence IDs; it cannot prove that a reply is semantically correct. Earlier evidence-overlap results mean that a gold section appeared in the retrieved set, not that the model cited that section correctly.

What we learned while building it

ISSUE	CHANGE
API quotas and service errors interrupted longer practice runs.	The question runner can resume completed responses, avoiding needless repeat calls after a quota interruption.
Embedding calls first failed on an unsupported request parameter and later hit quota limits.	The retriever falls back to lexical search when embeddings are unavailable; the final recorded hybrid run completed successfully.
The installed Gnani SDK rejected the first Hinglish TTS language setting.	The tested Hinglish configuration now uses `auto` for TTS. English, Hindi, and Hinglish sample replies produced audio.

Where the project stands

The core path is in place: retrieve policy, combine it with structured facts when needed, handle a conversation safely, and return speech through Gnani. The latest code changes also ask the model to name the evidence that supports its reply, validate those IDs against the material it received, and write them as `evidence_ids_used`. A per-file language manifest now lets one batch contain English, Hindi, Hinglish, or other Gnani locales supported by the configured account. These latest changes have not yet been rerun through the practice evaluations, so the scores above describe earlier runs.

Limits and next steps

NEXT	WHY IT MATTERS
Re-score after the evidence-ID change.	This will show how often the model selects the right supporting sections, rather than only whether retrieval found them.
Review a mixed-language batch and fresh recordings.	The language manifest is explicit rather than automatic, and current voice checks cover only a few self-recorded examples.
Record the required short demo and assemble the submission ZIP.	The demo should show a grounded answer, a follow-up, and a handover or an honest “I don’t know.” The ZIP must omit secrets, the dataset copy, and personal recordings.

Our main takeaway is simple: grounding improved the practice answers, while verification and explicit handovers kept the agent’s authority limited. The next useful measure is not another optimistic local score; it is a fresh run after the evidence changes, followed by a short demo that lets someone hear the system handle a real follow-up.

Public code: github.com/tanishka-td/Voxentra · Speech services: Gnani STT and TTS